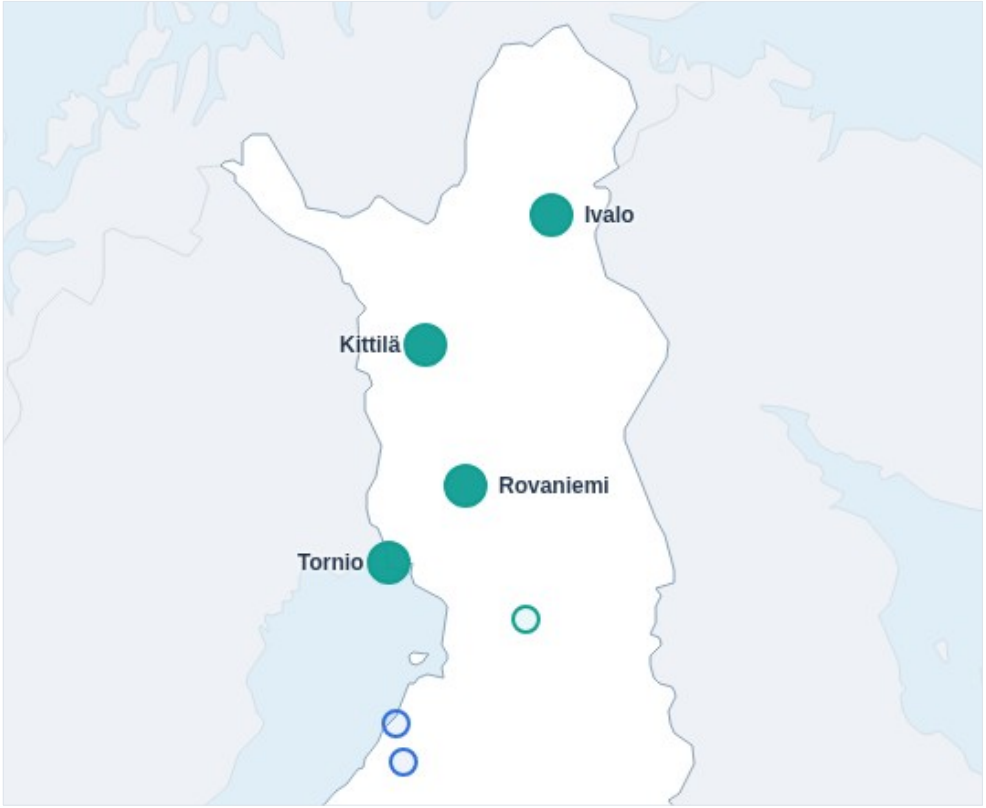
In May 2005, Kittilä and Ivalo flooded together.

The snow had warned us two months earlier.

1 March snow flagged 2005 in advance. Kittilä and Ivalo then passed their flood levels in the same spring.



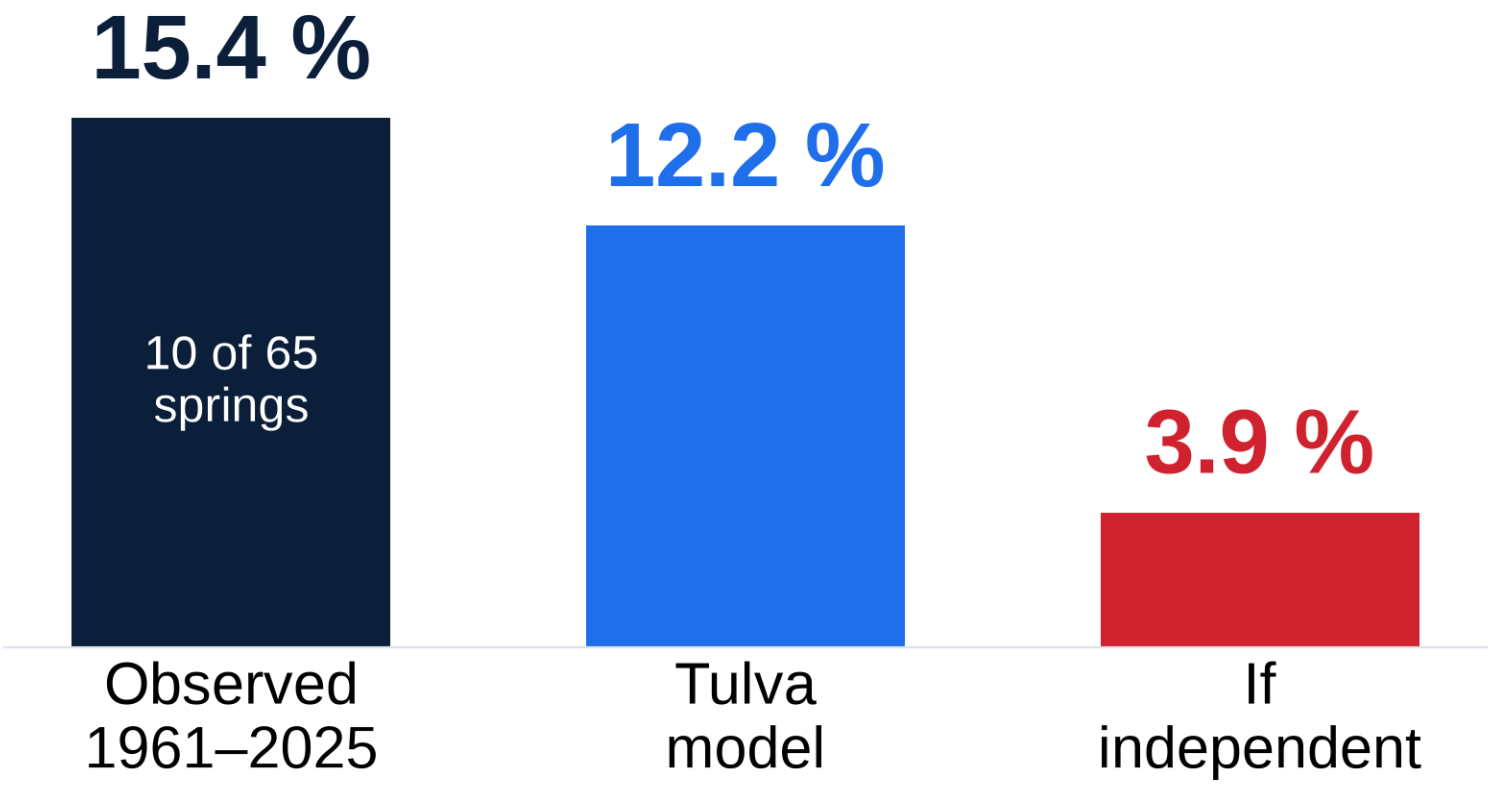
Tulva app, year replay: spring 2005

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

Finland's rivers flood together, not one at a time

How often do ≥4 of 14 river areas flood in the same spring?



Springs when ≥4 of the 14 river areas flooded (own 1-in-10 level)

15.4 %
observed, 1961–2025 (10 of 65 years)

12.2 %
Tulva joint model

3.9 %
if areas were independent

Home insurance pays only for floods rarer than 1-in-50.

Reinsurance renews 1 January, before the snow signal.

Source: SYKE 1961–2025 (+ SMHI Torne), Tulva model (exact); vesi.fi / OP (1-in-50 rule)

THE PRODUCT

Tulva Index: bought in winter, pays in spring

Pays €10M if ≥2 of 5 northern gauges pass their 1-in-50 level this spring

MARKET PRICE PER YEAR

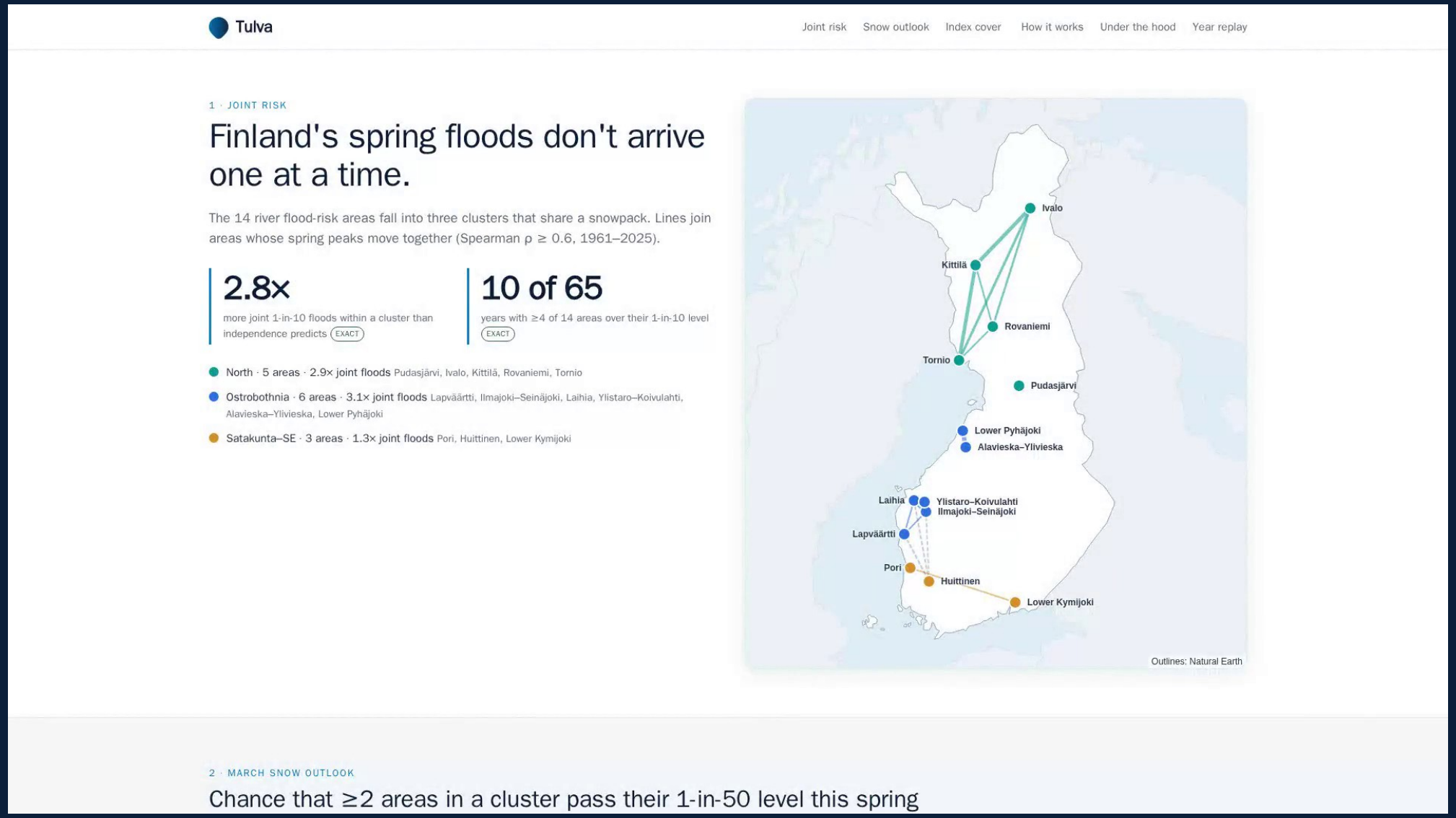


Bought in winter
before the snow signal exists

Staged option
50 % paid at 2 gauges, 100 % at 3

Public-gauge trigger
SYKE gauges anyone can check

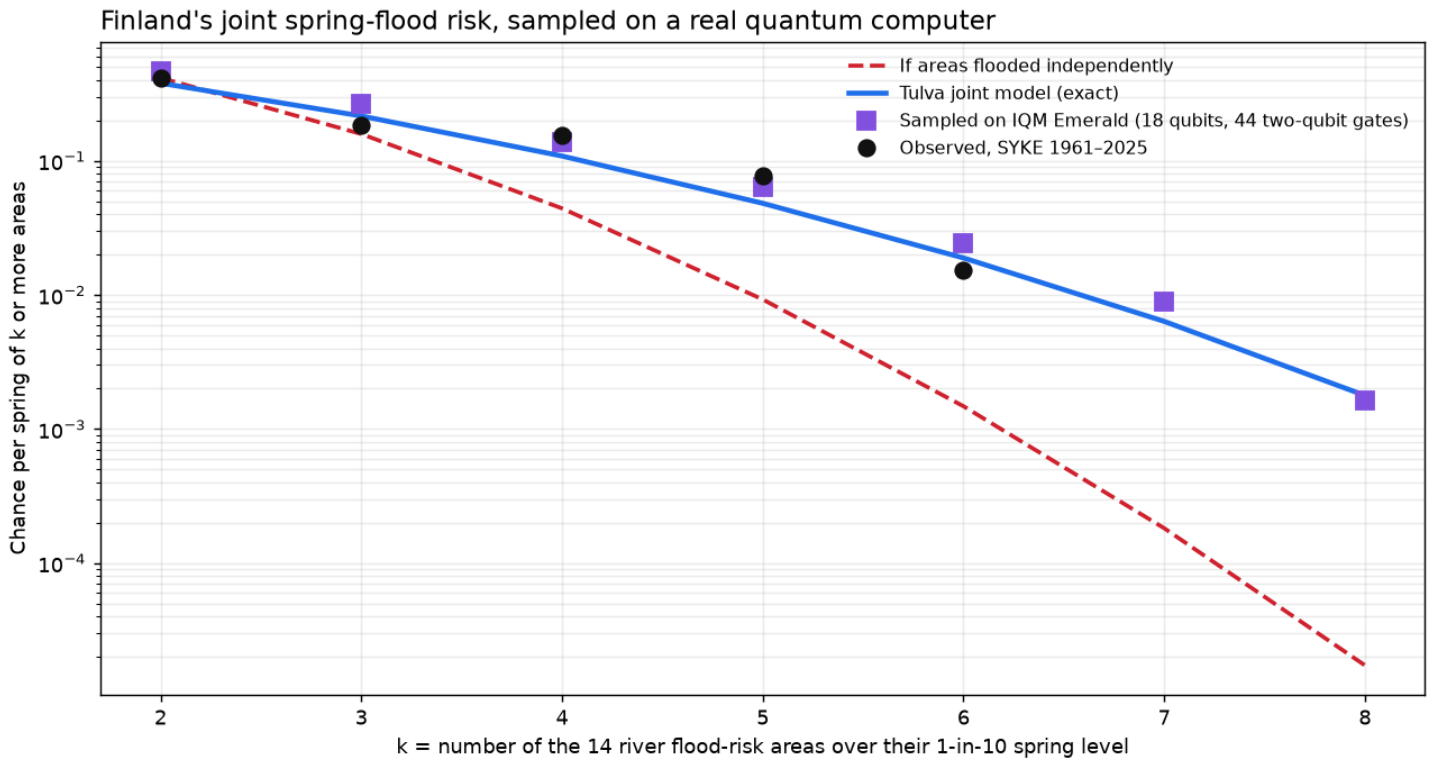
Source: SYKE 1961–2025, Tulva model (exact; MC 1M agrees). Market price = 2.5–3.5× expected payout (AM Best cat bonds, flood ILS)



Live app: the joint-flood map → March snow outlook → index price → real quantum runs → 2005 replay

QUANTUM, IN PRACTICE

The full model fits on a chip today; amplitude estimation works on a 3-qubit piece of it.



IQM Emerald via IQM Resonance, job 01a11cd4-da38-718b-80c9-7e259c6e4452, 8,000 shots, readout-corrected. Exact = two-level factor model fitted to SYKE spring peaks.

At the 1-in-10 flood level, the chip reproduces the joint tail that independence misses.

ALL 14 AREAS ON IQM EMERALD · 1-IN-10 LEVEL

18 qubits

44 two-qubit gates

AMPLITUDE ESTIMATION, IBM & IQM

P(Kittilä and Ivalo both pass 1-in-50)

Exact		**0.125 %**
IBM Marrakesh	0.262 →	**0.148 %**
IQM Garnet	0.368 →	**0.163 %**

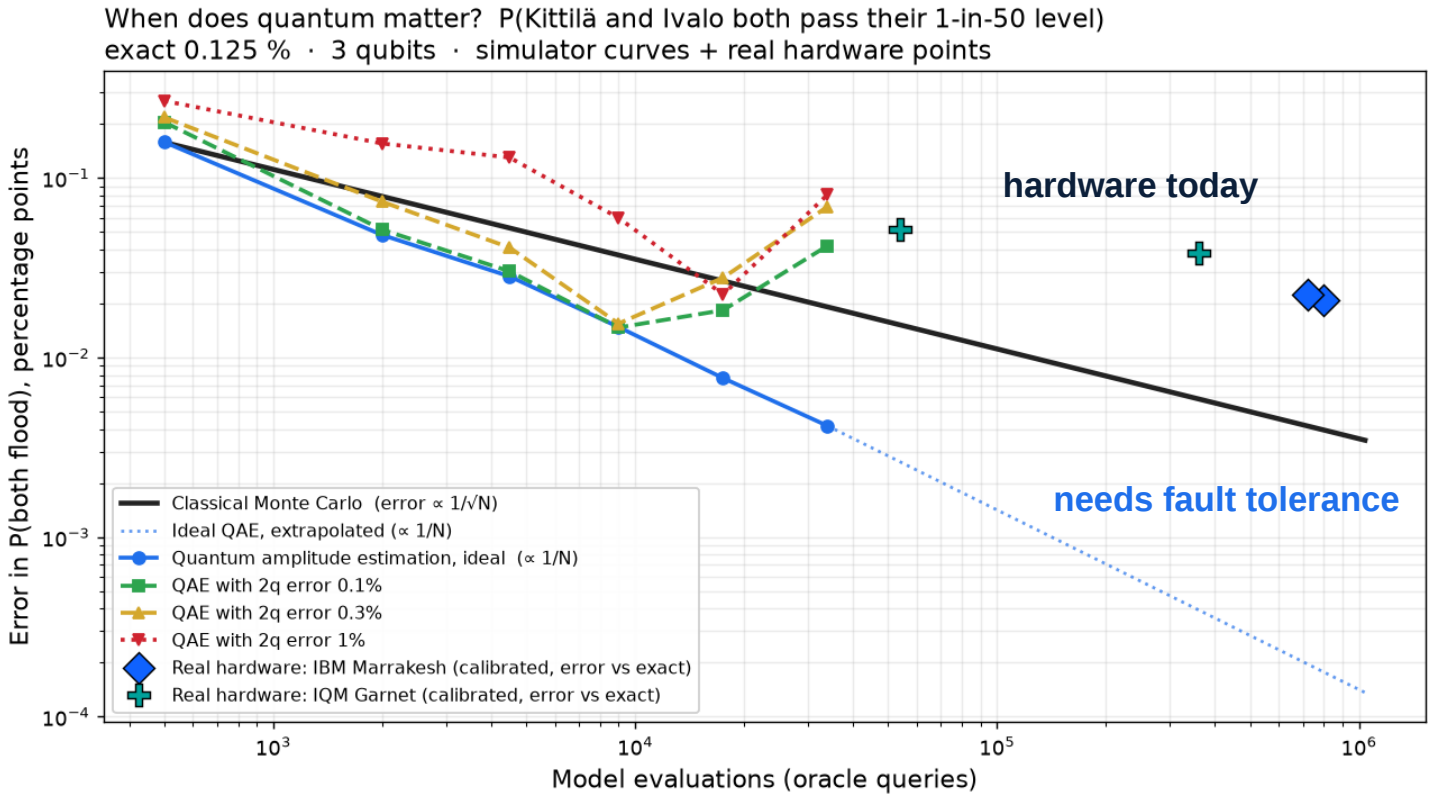
raw → calibrated

25 side-by-side copies on IBM Marrakesh · best-qubit selection from live calibration

Snow signal on the chip: **snowy vs low March 3.1× (exact 2.7×)**

IQM Emerald job 01a11cd4…; snow: Emerald, 10 copies × 240k shots; ibm_marrakesh 25 copies (table: 20 best-qubit copies); IQM Garnet 4 copies

Classical Monte Carlo still wins today



Fitted to SYKE spring peaks 1961–2025. Noisy curves: depolarising noise + 1 % readout, square-lattice routing; simulator error = RMS over 200 repetitions.
Hardware points = |calibrated estimate − exact| of one run (mostly residual bias), at that run's total model evaluations.

Where quantum could pay off

Fault-tolerant hardware
~8k logical qubits for comparable pricing (Chakrabarti et al. 2021)

Expensive models
full hydrological simulation per scenario

More work per price
multi-peril, daily repricing

Ideal QAE error falls like 1/N, Monte Carlo like 1/√N.

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

Who buys, why, and why now

WHO BUYS

Finnish non-life insurers

For the flood they keep under reinsurance. Four groups write 91 % of the market. Then reinsurers and ILS investors.

HOW WE EARN

Engine licence

Pricing and trigger engine, licensed per insurer or reinsurer. Structuring fee on placed covers.

WHY NOW

30 Jan 2027

Solvency II flood factor for Finland: 0 % correlation with SE/NO/DK (25 % for snowmelt rejected). ORSA needs climate scenarios.

SCALE

Finland → Nordics → Central Europe

Same engine, any snowmelt or river region. Gävle 2021: SEK 1.9bn claims. Storm Boris 2024: €2.15bn insured.

Ask **First step: a paid pilot with one Finnish insurer this autumn, feeding the 1 January reinsurance buy and ORSA.**

Sources: Finance Finland 2024; Del. Reg. (EU) 2026/269; EIOPA-BoS-24/462 §4.70; Finansinspektionen 2023 (Gävle); PERILS 2025 (Boris)

Tulva prices Finland's spring floods as a joint risk, and runs it on real quantum hardware.

BUILT THIS WEEKEND

- SYKE data pipeline
- Joint flood model
- Index pricing
- QAE on IBM & IQM

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack